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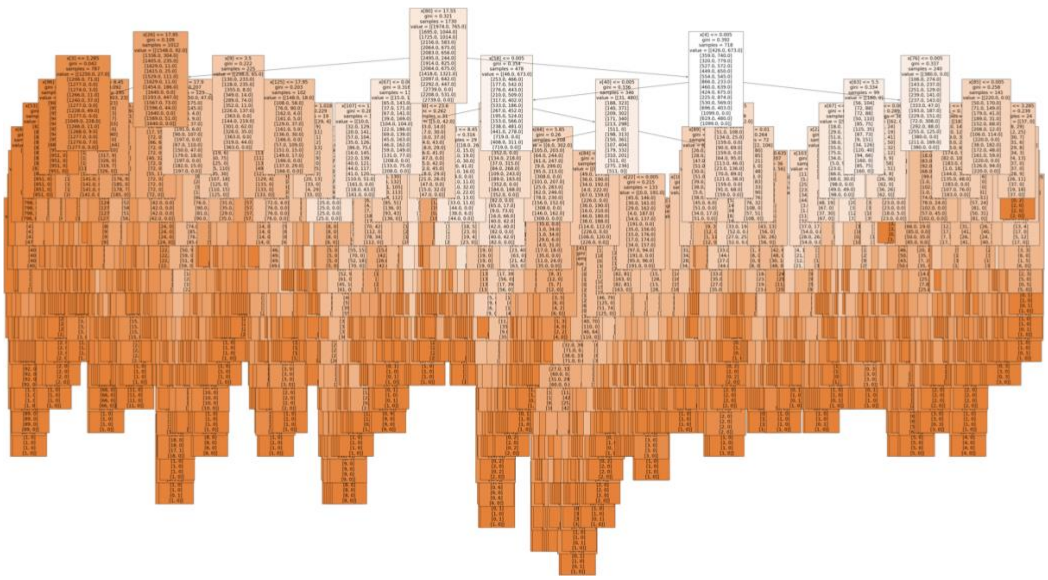
Exercise 2.4: Evaluating Hyperparameters for Model Optimization

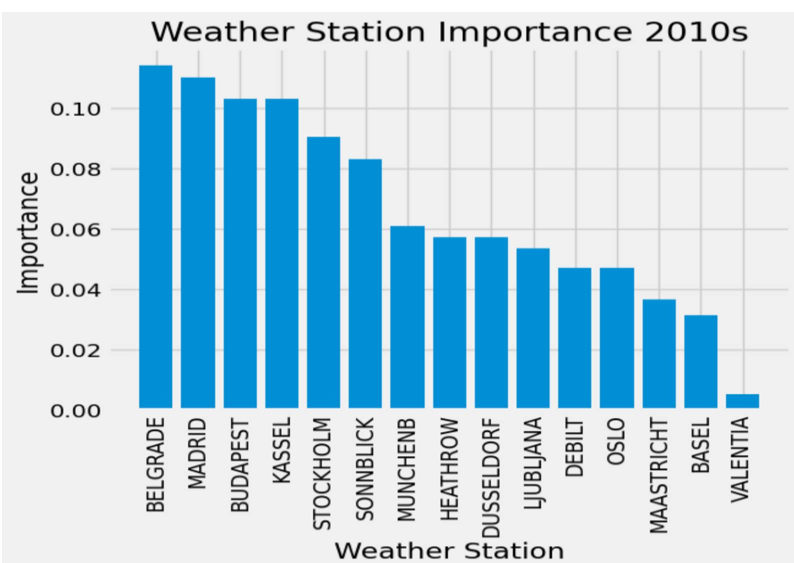
Part 1

Random Forest

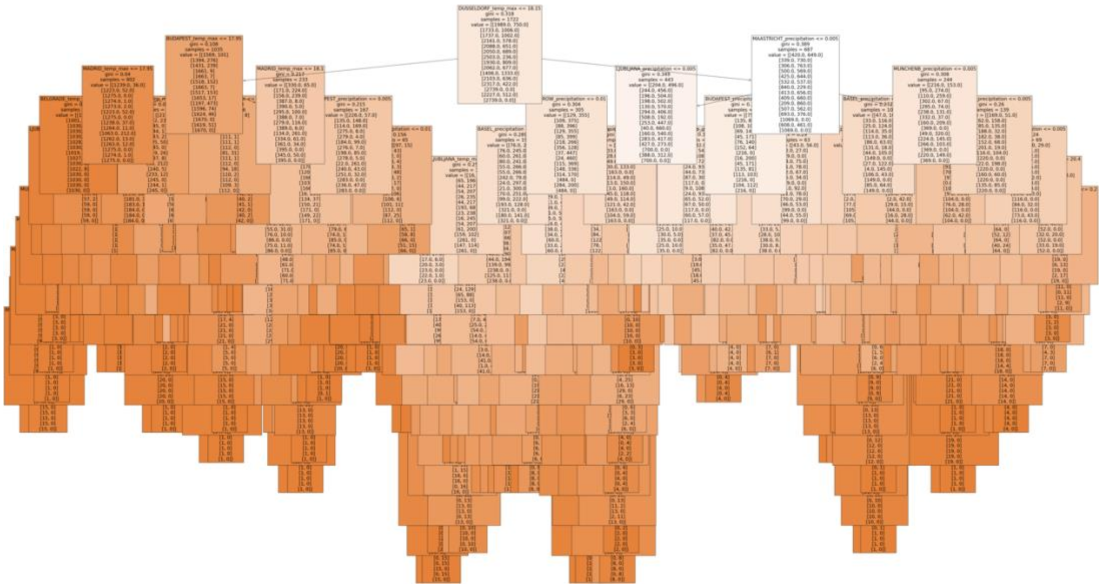
Datasets	Accuracy before Optimization	Accuracy after Optimization
All Weather Stations (2010 – 2019)	58.7%	65.27%
Stockholm(2010-2019)	99.98%	100%

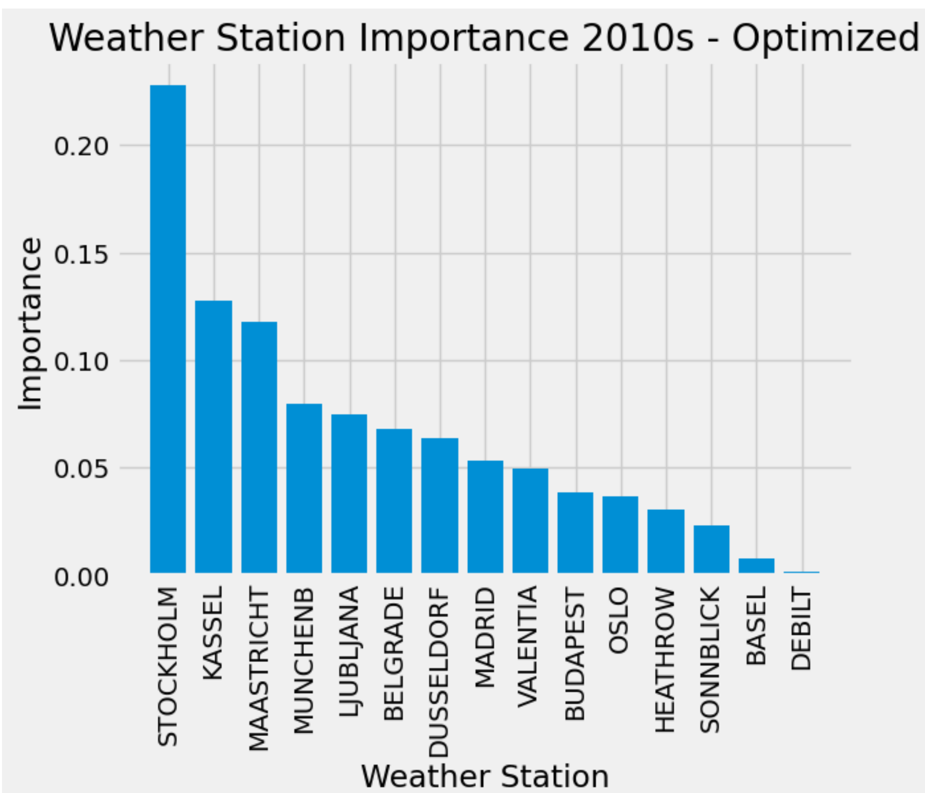
All Weather Stations (2010 – 2019) before Optimization.





All Weather Stations After Optimization

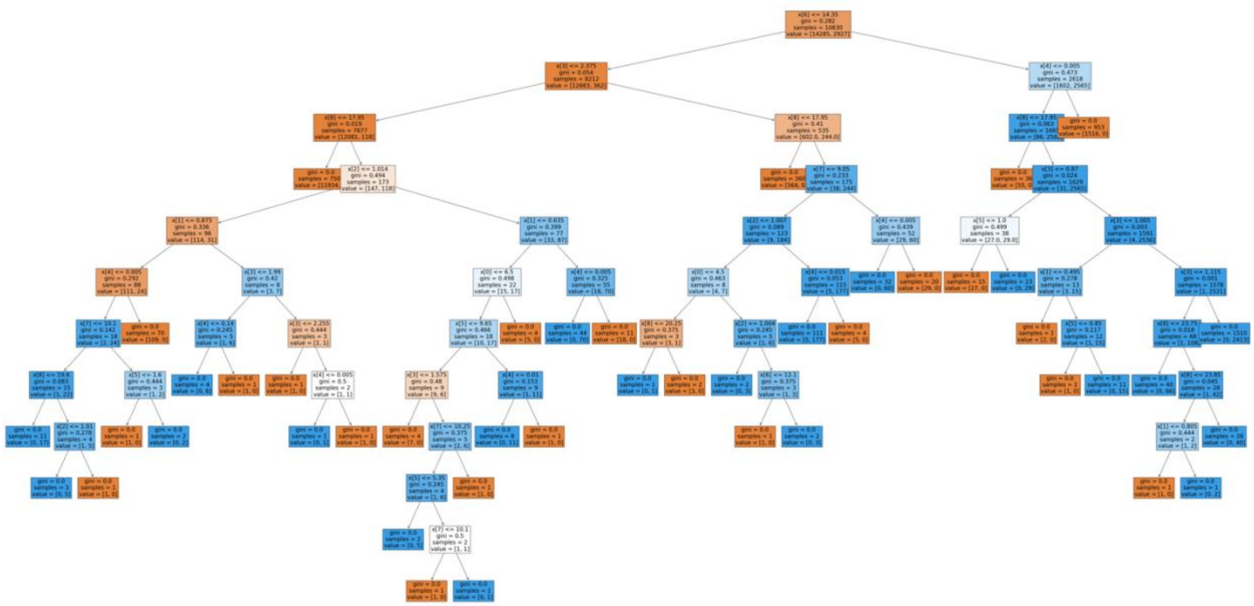


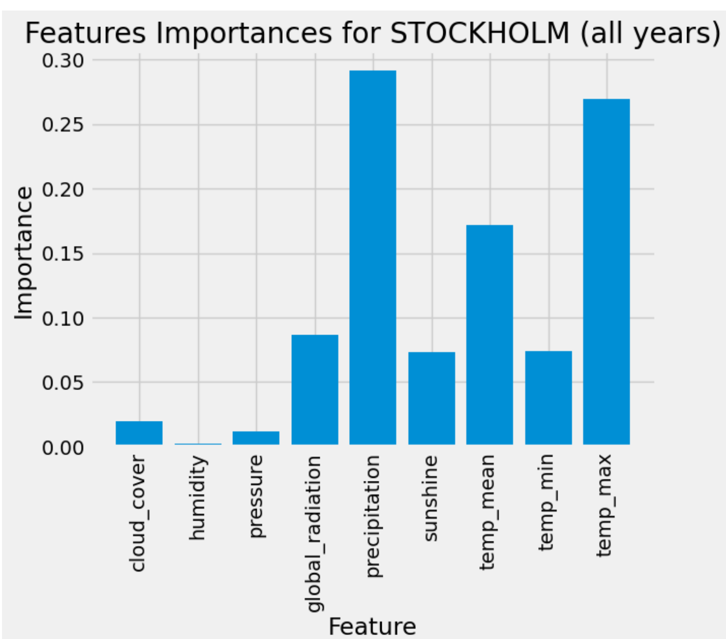


Model Accuracy: The model is improved by 6.5% after its optimized.

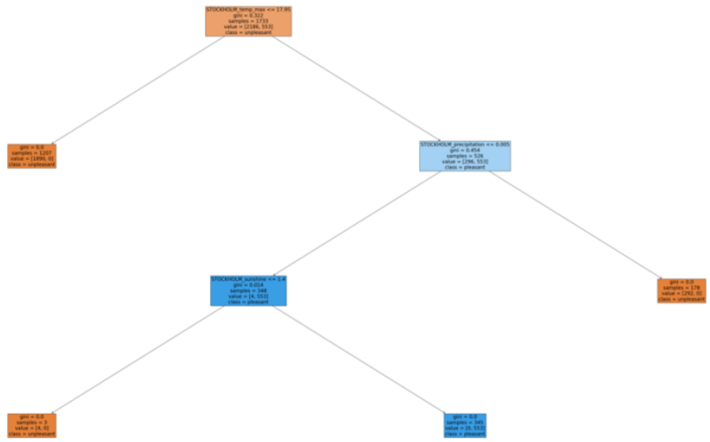
This decision tree is highly complex, with many layers and branches, indicating that it is fitting a large amount of detail from the training data. The top splits are based on key features like maximum temperature and precipitation from important stations such as Madrid, Budapest, and Dusseldorf, reflecting the model's focus on critical weather variables. However, the tree's deep structure also suggests a risk of overfitting if not properly controlled, even though it captures intricate patterns in the dataset.

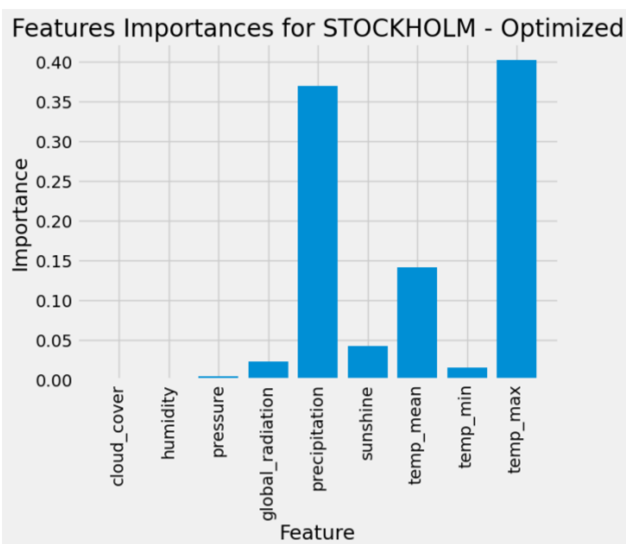
STOCKHOLM Before Optimization





STOCKHOLM After Optimization





After optimizing the hyperparameters for Stockholm, the model maintained a stable accuracy of 100%. The structure of the decision trees changed noticeably: the optimized model exhibits a simpler and more streamlined tree, which enhances generalization, interpretability, and overall performance. In contrast, the unoptimized model is more complex, increasing the risk of overfitting by capturing unnecessary noise from the data. Additionally, the importance of key features, such as maximum temperature and precipitation, shifted, with less relevant features now carrying even lower weights. This indicates that the optimized model is more efficient, concentrating on the most significant factors for accurately predicting weather conditions in Stockholm.

Part 2

Deep Learning

After optimization, the model's performance significantly improved. Accuracy rose from 11.87% to 92.64%, and the loss dropped from 53,369,332 to 0.2175, indicating better predictions and minimized errors. Although the confusion matrix now covers 12 out of 15 stations (compared to 13 before), the overall quality has greatly improved. The high accuracy, low loss, and focused feature importance suggest the model generalizes well without overfitting.

Hyperparameters:

Parameters	**Before Optimization**	**After Optimization**
Neurons/ N Hidden	256	61
Epochs	30	47
Batch Size	32	460
Optimizer	Adam	Adadelta
Kernel	2	2
Activation	relu	softsign
Learning Rate	-	0.763
Layers 1	-	1
Layers 2	-	2
Normalization	-	0.77
Dropout	-	0.72
Dropout rate	-	0.19

Pred True	BASEL	BELGRADE	BUDAPEST	DEBILT	DUSSELDORF	HEATHROW	KASSEL \
BASEL	3379	157	23	10	12	14	2
BELGRADE	48	1037	2	1	0	1	0
BUDAPEST	18	26	163	6	1	0	0
DEBILT	7	4	11	54	4	2	0
DUSSELDORF	3	1	1	8	12	4	0
HEATHROW	5	3	3	2	6	63	0
KASSEL	0	3	1	0	1	0	3
LJUBLJANA	5	5	3	0	1	5	1
MAASTRICHT	2	1	0	1	0	1	0
MADRID	12	13	21	1	4	25	0
MUNCHENB	5	2	0	0	0	0	0
OSLO	0	0	0	0	1	0	0
STOCKHOLM	0	0	1	0	0	1	0
VALENTIA	1	0	0	0	0	0	0

Pred True	LJUBLJANA	MAASTRICHT	MADRID	MUNCHENB	OSLO	VALENTIA
BASEL	12	1	71	0	0	1
BELGRADE	2	0	1	0	0	0
BUDAPEST	0	0	0	0	0	0
DEBILT	0	0	0	0	0	0
DUSSELDORF	0	0	0	0	0	0
HEATHROW	0	0	0	0	0	0
KASSEL	2	1	0	0	0	0
LJUBLJANA	40	0	1	0	0	0
MAASTRICHT	0	4	0	0	0	0
MADRID	10	0	372	0	0	0
MUNCHENB	0	0	0	1	0	0
OSLO	0	0	0	0	4	0
STOCKHOLM	0	0	0	0	2	0
VALENTIA	0	0	0	0	0	0

The optimization of the model led to a substantial improvement in performance, with the accuracy increasing by 80.28%, going from a relatively low 11.9% to a much higher 92.2%. This signifies a major enhancement in the model's ability to make correct predictions or classifications.

Part 3

Iteration

To optimize weather predictions for the Air Ambulance company, it is advisable to segment the dataset into smaller, more manageable components based on factors such as location, time intervals, and specific weather features. For location-based segmentation, the data can be divided by individual weather stations or groups of stations with similar weather patterns, allowing the model to focus on local weather trends. Additionally, segmenting the data by time periods such as seasons, months, or years could help capture seasonal variations, while grouping data by specific weather metrics such as temperature, precipitation, and sunshine would highlight how each factor contributes to flight safety.

In terms of model selection, both Random Forest and Convolutional Neural Networks (CNNs) have distinct advantages and limitations. Random Forest, being an optimized model, offers transparency and interpretability, making it easier to understand and explain the predictions. It requires fewer hyperparameters to tune compared to deep learning models, and its ensemble approach helps reduce overfitting by averaging the results from multiple trees, leading to more reliable and generalizable predictions. However, it has limitations in capturing complex, non-linear relationships and may not generalize well to new time periods outside the training data. Additionally, it can become computationally expensive when dealing with large datasets. On the other hand, CNNs excel at identifying intricate patterns, particularly in large datasets, and are capable of learning directly from raw data, which reduces the need for extensive feature engineering. They also scale effectively with increasing data volumes. However, CNNs lack transparency, making them more difficult to interpret, and they require significant computational resources and careful hyperparameter tuning to avoid overfitting.

The key variables for the Air Ambulance to consider when assessing flight safety:

- Weather Conditions: This includes visibility, wind speed and direction, temperature, precipitation, and the presence of thunderstorms, all of which directly affect flight safety.
- Aircraft Readiness: Ensuring the aircraft is in good condition, for instances, engine performance, fuel levels, and systems functionality.
- Air Traffic Control: Monitoring the flight route for air traffic restrictions, hazardous terrain, and ensuring safe navigation corridors are clear is essential for safety.
- Landing Site Conditions: Evaluating the integrity of the landing area (runway or helipad), its altitude, and potential obstacles ensures a safe and successful landing.